

Seven Seas A-35

Security Audit

March 25, 2025

Version 1.0.0

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Introduction

This document includes the results of the security audit for Seven Seas's smart contract code as found in the section titled 'Source Code'. The security audit was performed by the Macro security team over multiple days starting on March 18th, 2025.

The purpose of this audit is to review the source code of certain Seven Seas Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro's review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

The following is an aggregation of issues found by the Macro Audit team:

Severity	Count	Acknowledged	Won't Do	Addressed
Code Quality	2	-	-	2

Seven Seas was quick to respond to these issues.

Specification

Our understanding of the specification was based on the following sources:

- Discussions with the Seven Seas team.
- Available documentation in the repository.

Source Code

The following source code was reviewed during the audit:

- **Odos Decoder (from [PR 184](#))**

Commit Hash: `f472abafb22f7a45e81df2814ae4bdf6f56dd74a`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/Protocols/OdosDecoderAndSanitizer.sol</code>	<code>99df2d0b09be291f4eee4857ad21fce43cf6540ed7c6231bb10197564688581b</code>

- **Elixir Decoder (from [PR 185](#))**

Commit Hash: `187e7b0b619dfdcbe78273d41b7fe24d1bb22ed7`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/Protocols/ElixirClaimingDecoderAndSanitizer.sol</code>	<code>1b8ccee10b662f9d8c31e021048e95a85fe18de0ec4fddd9774a14e1828b897</code>

- **Syrup Decoder (from [PR 179](#))**

Commit Hash: `e80c7df3607fb8e84fd323720aed23f98213227a`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/Protocols/SyrupDecoderAndSanitizer.sol</code>	<code>bf432c48e72ee1d0d77884b5d8184c0a05a662c4e2d0186d3032eeffb9615ab1</code>

- **BalancerV3 Decoder (from [PR 190](#))**

Commit Hash: `8530fbe7225fb699a1924aa29545f17c622f28e4`

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/BalancerV3DecoderAndSanitizer.sol	e4d359e1b6b03852bfb524de6d5acf01019e214f9083790272142ba129401a73
src/base/DecodersAndSanitizers/Protocols/Permit2DecoderAndSanitizer.sol	d30a5067007016755a14f5f8320a68deb7fcbd45896593d07c08dd808e5dab7

- **Royco Decoder (from [PR 188](#))**

Commit Hash: 8530fbe7225fb699a1924aa29545f17c622f28e4

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/RoycoDecoderAndSanitizer.sol	2cbc4b13b799fb4a26c23cd40648b4c2144dd0e2d3626e583f96ef3f9938169c

Note: This document contains an audit solely of the Solidity contracts listed above. Specifically, the audit pertains only to the contracts themselves, and does not pertain to any other programs or scripts, including deployment scripts.

Issue Descriptions and Recommendations

Click on an issue to jump to it, or scroll down to see them all.

[Q-1](#) Permit2 decoder doesn't support `lockdown`

[Q-2](#) Royco decoder doesn't support cancelling offers on the `VaultMarketHub`

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:

- How bad things can get (for a vulnerability)
- The significance of an improvement (for a code quality issue)
- The amount of gas saved (for a gas optimization)

2. The high/medium/low **likelihood** of the issue:

- How likely is the issue to occur (for a vulnerability)

3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client must fix the issue, no matter what, because not fixing would mean significant funds/assets WILL be lost.
(H-x) High	We recommend the client must address the issue, no matter what, because not fixing would be very bad, or some funds/assets will be lost, or the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to seriously consider fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albeit not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not be relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Details

Q-1

Permit2 decoder doesn't support Lockdown

TOPIC	STATUS	QUALITY IMPACT
Missing Functionality	Fixed ↗	Medium

For the Permit2DecoderAndSanitizer , it could be useful to allow revoking approvals by supporting the lockdown function.

Remediation to Consider Add the lockdown function to the Permit2DecoderAndSanitizer.

Q-2

Royco decoder doesn't support cancelling offers on the VaultMarketHub

TOPIC	STATUS	QUALITY IMPACT
Missing Functionality	Fixed ↗	Medium

The RoycoDecoderAndSanitizer allows creating AP (action provider) offers on both the RecipeMarketHub and VaultMarketHub. While offer cancellation is implemented for RecipeMarketHub through cancelAPOffer , no equivalent cancellation function exists for VaultMarketHub.

Remediation to Consider Add support for cancelling offers on the VaultMarketHub by implementing the cancelOffer function.

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The scope of this report and review is limited to a review of only the code presented by the Seven Seas team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.